$2f_A(x)\cdot f_B(x)=0$.

$\therefore 1=1+0-0 \Rightarrow \boxed{f_{A\triangle B}(x)=f_A(x)+f_B(x)-2f_A(x)\cdot f_B(x)}$

Let, $x\notin A\triangle B$. By defn. of characteristic function, $f_{A\triangle B}(x)=0$.
By Defn. of symmetric difference, x either belongs to both or belongs to neither.

① $x\in A, x\in B$. By defn. of characteristic function, $f_A(x)=f_B(x)=1$.

$f_A(x)\cdot f_B(x)=1 \quad \therefore 0=2-2=2-2\cdot 1=1+1-2\cdot 1$

$\Rightarrow \boxed{f_{A\triangle B}(x)=f_A(x)+f_B(x)-2f_A(x)\cdot f_B(x)}$

② $x\notin A, x\notin B$. By defn. of characteristic function, $f_A(x)=f_B(x)=0$.

$f_A(x)\cdot f_B(x)=0 \quad \therefore 0=0+0-2\cdot 0 \Rightarrow \boxed{f_{A\triangle B}(x)=f_A(x)+f_B(x)-2f_A(x)\cdot f_B(x)}$

□

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74) Suppose that f is a function from A to B, where A and B are finite sets with $|A|=|B|$. Show that f is one-to-one if and only if it is onto.

Soln: Given, A and B are finite sets, $|A|=|B|$.

Let, $|A|=|B|=m$.

Let, f be one-to-one. By Defn. 5, if $\forall a,b\in A, f(a)=f(b)$, then $a=b$.

Let, f be non-onto. $\therefore \exists x\in B$ s.t. $\forall y\in A, f(y)\neq x$. (By Defn. 7)

$\therefore$ The m elements of A must be mapped with the $m-1$ elements of B apart from element $x\in B$.

By Defn. 1, each element of A is mapped to a unique element of B.

$\therefore$ There must exist two elements $p,q\in A$ s.t. $f(p)=f(q)$, $p\neq q$.

This contradicts the fact that f is one-to-one.

$\therefore f$ must be onto.

Let, f be onto. By Defn. 7, $\forall y\in B, \exists x\in A$ s.t. $f(x)=y$.

By Defn. 1, each element of A is mapped to a unique element of B.

$\therefore |A|\geq|B|$. (Is it rigorous?). Given $|A|=|B|$.

Let, f not be one-to-one. By Defn. 5, there exists two elements $p,q\in A$, $p\neq q$ s.t. $f(p)=f(q)$.